A SPECTRAL BIJECTION LIFTING SPECIALIZATIONS OR GENERALIZATIONS IS A HOMEOMORPHISM

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that the constructible topology of a spectral space is Hausdorff and quasi-compact. Invoked inside the counted proof (source lines 4835-4836). Puts the map in the setting where a bijective map is a homeomorphism.
- The statement that a bijective map of quasi-compact Hausdorff spaces is a homeomorphism. Invoked inside the counted proof (source line 4836). Gives the homeomorphism in the constructible topology.
- The statement that a constructible subset stable under specialization (or generalization) is closed, respectively open. Invoked inside the counted proof (source lines 4844-4845). Concludes that the image of a quasi-compact open is open.

Source reference. Original lemma lemma-bijective-map: sources/stacks-topology/pinned-original-source.tex, source lines 3498–3498. Navigation pointer only: the counted text refers to the Stacks item lemma-bijective-map, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-constructible-hausdorff-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 4522–4522. Navigation pointer only: the counted text refers to the Stacks item lemma-constructible-hausdorff-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

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Source reference. Original lemma lemma-constructible-stable-specialization-closed: sources/stacks-topology/pinned-original-source.tex, source lines 4678–4678. Navigation pointer only: the counted text refers to the Stacks item lemma-constructible-stable-specialization-closed, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $f : X \to Y$ be a continuous map of topological spaces. If*

(1) *X and Y are spectral,*
(2) *f is spectral and bijective, and*
(3) *generalizations (resp. specializations) lift along f.*

Then f is a homeomorphism.

Proof. Since f is spectral it defines a continuous map between X and Y in the constructible topology. By Lemmas lemma-constructible-hausdorff-quasi-compact and lemma-bijective-map it follows that $X \to Y$ is a homeomorphism in the constructible topology. Let $U \subset X$ be quasi-compact open. Then $f(U)$ is constructible in Y. Let $y \in Y$ specialize to a point in $f(U)$. By the last assumption we see that $f^{-1}(y)$ specializes to a point of U. Hence $f^{-1}(y) \in U$. Thus $y \in f(U)$. It follows that $f(U)$ is open, see Lemma lemma-constructible-stable-specialization-closed. Whence f is a homeomorphism. To prove the lemma in case specializations lift along f one shows instead that $f(Z)$ is closed if $X \setminus Z$ is a quasi-compact open of X. □